Epoch Ledger

Every completed epoch so far, stage 1 through stage 3 epoch 10 - loss, ordinal agreement, minority-class recall, screening/STDR discrimination, hard-example-mining state, and the full per-grade recall spread, read directly from the training log.

EPOCHS COMPLETE	BEST QWK	BEST SCORE	BEST MIN-CLASS RECALL	BEST REFAUC
28 / 45	0.607	0.505	0.474	0.873
Stage 3 epoch 12 in progress	Stage 3, epoch 6 (was 0.554)	Stage 3, epoch 6 (was 0.471)	Stage 2, epoch 7	Stage 3, epoch 6 (was 0.848)

Column glossary

Time — context

Wall-clock minutes to train the epoch; not a quality signal.

Loss — lower better

Weighted sum of every training objective, averaged per image.

QWK — higher better

Quadratic Weighted Kappa: ordinal agreement, penalises distant misses harder.

F1 — higher better

Macro F1 across all 5 grades, treating each equally.

Min-recall — higher better

Recall of whichever grade is doing worst that epoch - the key imbalance metric.

Score — higher better

Composite selection score used to pick the best checkpoint.

Accuracy — higher, cautiously

Overall correct-grade rate; can rise just by favouring grade 0.

refAUC — higher better

AUC for the binary referable-vs-not-referable DR split.

Screen cleared — context

Share of images passed without flagging; trades off against NPV.

Screen NPV — higher better

Negative predictive value of the screening gate vs a 0.985 target.

Screen thr — context

Decision threshold that hit the NPV target this epoch.

STDR AUC — higher better

AUC for detecting sight-threatening DR (Severe + PDR).

Miner p90 H — context

90th-percentile mining hardness carried into the next epoch.

Boundary rate — context

Share of samples flagged as one-grade boundary errors.

Per-grade recall — higher better, all 5

Recall per grade 0-4 - see the dedicated heatmap page.

Full comparison, all 28 epochs

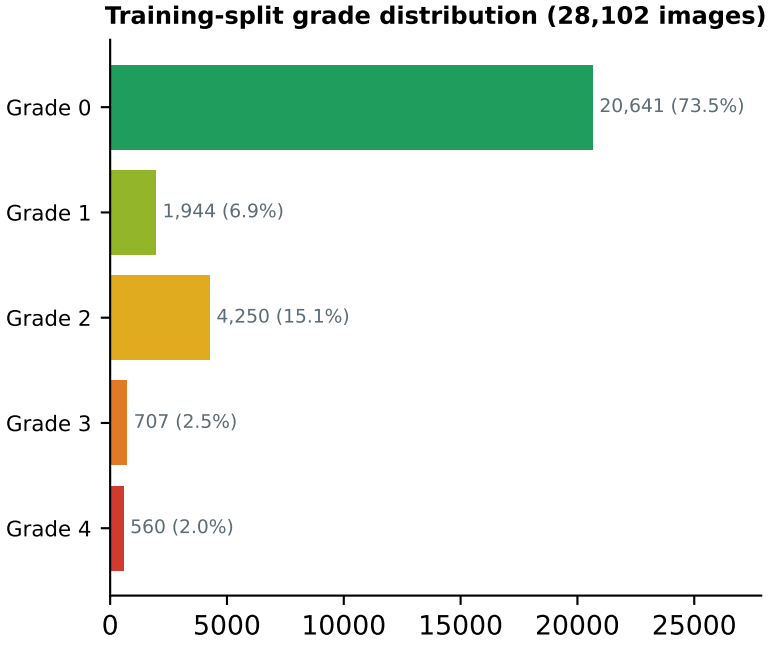
Stage 1: head only, backbone frozen. Stage 2: GLA-LoRA + ALPP trainable, backbone still frozen. Stage 3: 8/24 backbone blocks unfrozen too.
Shading: worst (red) to best (green) per column; loss inverted (lower is greener).

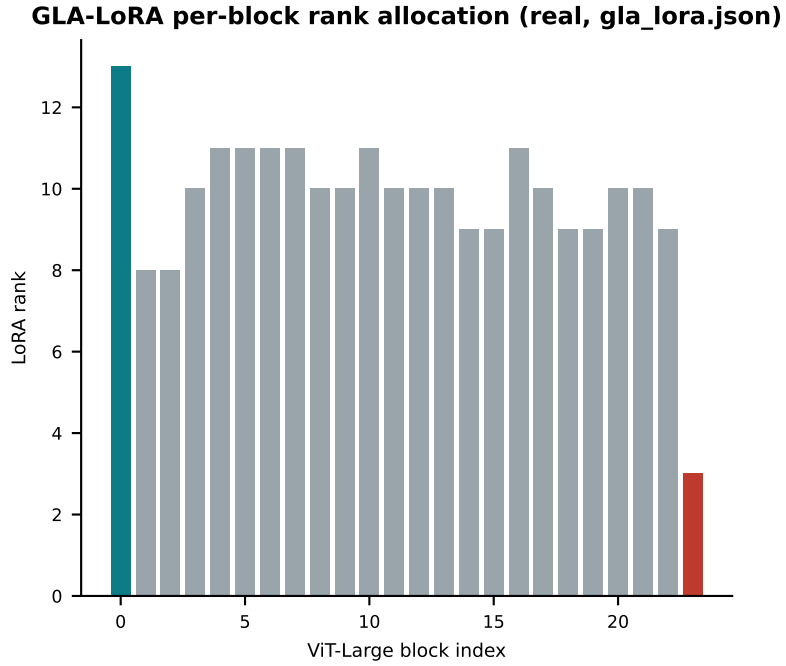
Epoch	Time	Loss	QWK	F1	Min-recall	Score	Accuracy	refAUC	Screen cleared	Screen NPV	Screen thr	STDR AUC	Miner p90 H	Boundary rate
S1 e1	75.7m	83.90	0.351	0.292	0.187	0.300	0.666	0.726	91.1%	0.9854	0.237	0.8877	1.00	0.03
S1 e2	68.8m	48.43	0.421	0.287	0.198	0.336	0.607	0.778	94.1%	0.9851	0.599	0.9162	1.00	0.05
S1 e3	75.6m	33.65	0.441	0.284	0.168	0.339	0.634	0.794	93.9%	0.9851	0.590	0.9165	1.15	0.06
S1 e4	75.6m	27.29	0.403	0.301	0.289	0.350	0.569	0.800	94.1%	0.9851	0.668	0.9149	1.28	0.07
S1 e5	70.1m	25.51	0.458	0.309	0.211	0.364	0.639	0.808	93.9%	0.9851	0.646	0.9176	1.37	0.08
S2 e1	83.4m	24.05	0.434	0.287	0.228	0.349	0.539	0.809	94.9%	0.9852	0.697	0.9203	1.43	0.08
S2 e2	95.0m	17.47	0.496	0.376	0.269	0.414	0.725	0.804	94.7%	0.9852	0.470	0.9163	1.47	0.08
S2 e3	85.7m	12.15	0.465	0.303	0.238	0.371	0.609	0.813	94.6%	0.9852	0.652	0.9201	1.52	0.08
S2 e4	98.8m	9.34	0.456	0.318	0.301	0.384	0.587	0.820	94.5%	0.9852	0.686	0.9215	1.55	0.09
S2 e5	100.8m	8.13	0.554	0.399	0.370	0.471	0.703	0.832	94.2%	0.9851	0.564	0.9461	1.58	0.09
S2 e6	91.5m	7.34	0.554	0.390	0.334	0.461	0.691	0.836	94.2%	0.9851	0.549	0.9345	1.59	0.09
S2 e7	98.4m	6.95	0.402	0.344	0.474	0.399	0.479	0.836	94.9%	0.9853	0.662	0.9459	1.61	0.09
S2 e8	91.0m	6.62	0.509	0.380	0.385	0.445	0.611	0.839	95.5%	0.9853	0.643	0.9413	1.61	0.09
S2 e9	96.6m	6.37	0.474	0.366	0.418	0.430	0.578	0.848	95.0%	0.9853	0.693	0.9426	1.60	0.09
S2 e10	83.5m	6.20	0.482	0.380	0.438	0.443	0.587	0.839	94.9%	0.9852	0.691	0.9410	1.60	0.09
S2 e11	92.2m	6.08	0.510	0.391	0.435	0.459	0.615	0.839	94.7%	0.9852	0.665	0.9499	1.61	0.09
S2 e12	101.9m	6.11	0.493	0.387	0.439	0.450	0.603	0.840	94.9%	0.9852	0.667	0.9461	1.61	0.09
S3 e1	212.7m	6.00	0.448	0.351	0.443	0.418	0.544	0.839	94.9%	0.9853	0.687	0.9549	1.61	0.09
S3 e2	194.1m	5.87	0.479	0.370	0.445	0.440	0.600	0.845	95.7%	0.9854	0.659	0.9532	1.61	0.09
S3 e3	214.3m	5.52	0.506	0.375	0.436	0.453	0.585	0.855	95.8%	0.9854	0.683	0.9613	1.61	0.09
S3 e4	217.1m	5.32	0.549	0.393	0.416	0.476	0.632	0.863	96.2%	0.9854	0.695	0.9645	1.60	0.10
S3 e5	222.1m	5.17	0.577	0.399	0.395	0.487	0.675	0.862	96.1%	0.9854	0.680	0.9657	1.60	0.10
S3 e6	302.4m	5.09	0.607	0.412	0.389	0.505	0.697	0.873	95.8%	0.9854	0.660	0.9628	1.61	0.10
S3 e7	14.0m	4.87	0.557	0.401	0.420	0.483	0.645	0.870	95.8%	0.9854	0.676	0.9640	1.61	0.10
S3 e8	48.3m	4.84	0.584	0.421	0.402	0.499	0.663	0.871	96.1%	0.9854	0.679	0.9622	1.61	0.10
S3 e9	32.9m	4.73	0.588	0.420	0.417	0.504	0.672	0.870	94.9%	0.9852	0.623	0.9593	1.61	0.10
S3 e10	116.6m	4.62	0.529	0.402	0.466	0.478	0.585	0.865	95.7%	0.9854	0.680	0.9576	1.60	0.10
S3 e11	226.2m	4.59	0.590	0.427	0.109	0.510	0.670	0.869	95.1%	0.9853	0.616	0.9595	1.60	0.10

Per-grade recall, every epoch

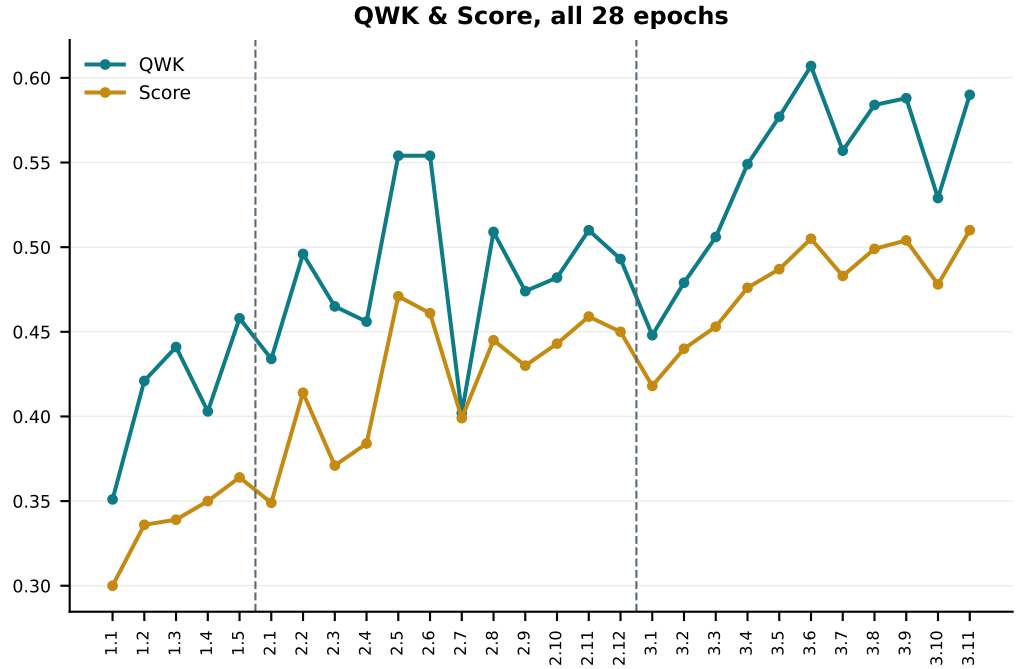
Shaded per grade (each column against its own best/worst across all 28 epochs) so a hard grade like Mild still shows its own real progress.

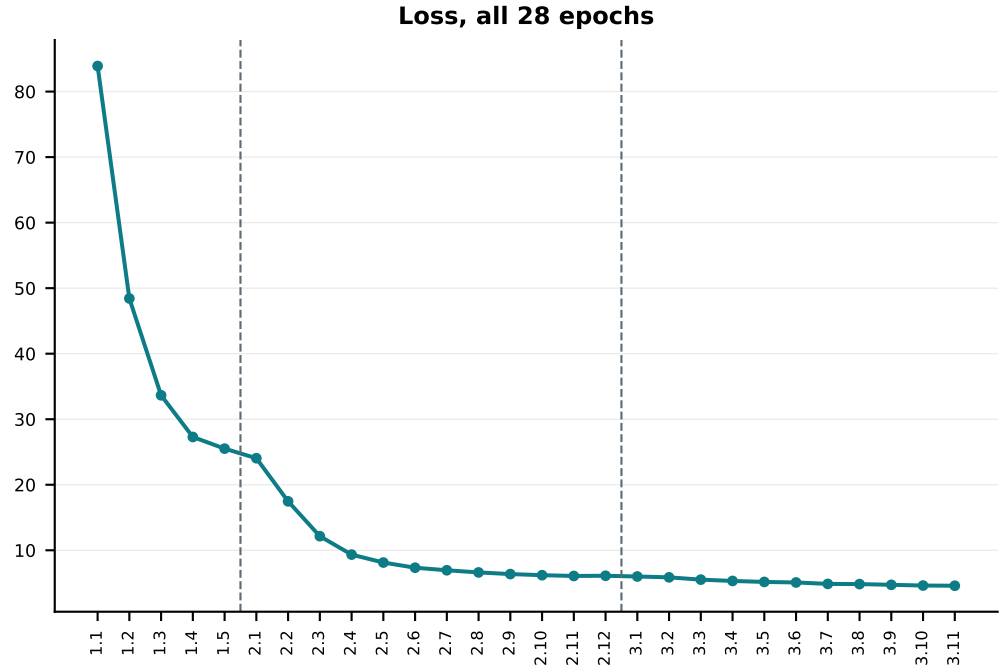
Epoch	Grade 0 No DR	Grade 1 Mild	Grade 2 Moderate	Grade 3 Severe	Grade 4 PDR
S1 e1	82.9%	5.9%	22.4%	27.8%	0.0%
S1 e2	72.6%	5.9%	34.1%	19.4%	28.6%
S1 e3	76.7%	6.9%	32.2%	11.1%	28.6%
S1 e4	65.3%	10.9%	39.7%	36.1%	33.3%
S1 e5	75.8%	5.9%	37.9%	19.4%	42.9%
S2 e1	61.1%	14.9%	42.5%	11.1%	42.9%
S2 e2	89.5%	5.0%	22.9%	52.8%	23.8%
S2 e3	71.9%	9.9%	36.4%	25.0%	28.6%
S2 e4	67.6%	10.9%	40.7%	38.9%	33.3%
S2 e5	83.5%	5.0%	36.4%	69.4%	23.8%
S2 e6	82.4%	5.0%	34.1%	61.1%	28.6%
S2 e7	48.9%	21.8%	53.7%	66.7%	28.6%
S2 e8	69.5%	9.9%	44.4%	61.1%	28.6%
S2 e9	63.9%	13.9%	47.7%	63.9%	33.3%
S2 e10	65.6%	12.9%	43.5%	75.0%	38.1%
S2 e11	69.5%	11.9%	43.5%	75.0%	33.3%
S2 e12	67.6%	12.9%	43.9%	75.0%	33.3%
S3 e1	60.2%	8.9%	43.5%	80.6%	28.6%
S3 e2	67.9%	11.9%	41.1%	80.6%	23.8%
S3 e3	65.9%	11.9%	41.1%	77.8%	28.6%
S3 e4	72.3%	10.9%	41.6%	72.2%	28.6%
S3 e5	79.1%	13.9%	35.0%	69.4%	28.6%
S3 e6	82.0%	10.9%	36.4%	69.4%	28.6%
S3 e7	73.8%	10.9%	43.0%	72.2%	33.3%
S3 e8	76.6%	10.9%	40.2%	69.4%	42.9%
S3 e9	77.9%	11.9%	38.3%	75.0%	38.1%
S3 e10	64.5%	21.8%	43.0%	75.0%	42.9%
S3 e11	77.0%	10.9%	41.6%	77.8%	38.1%

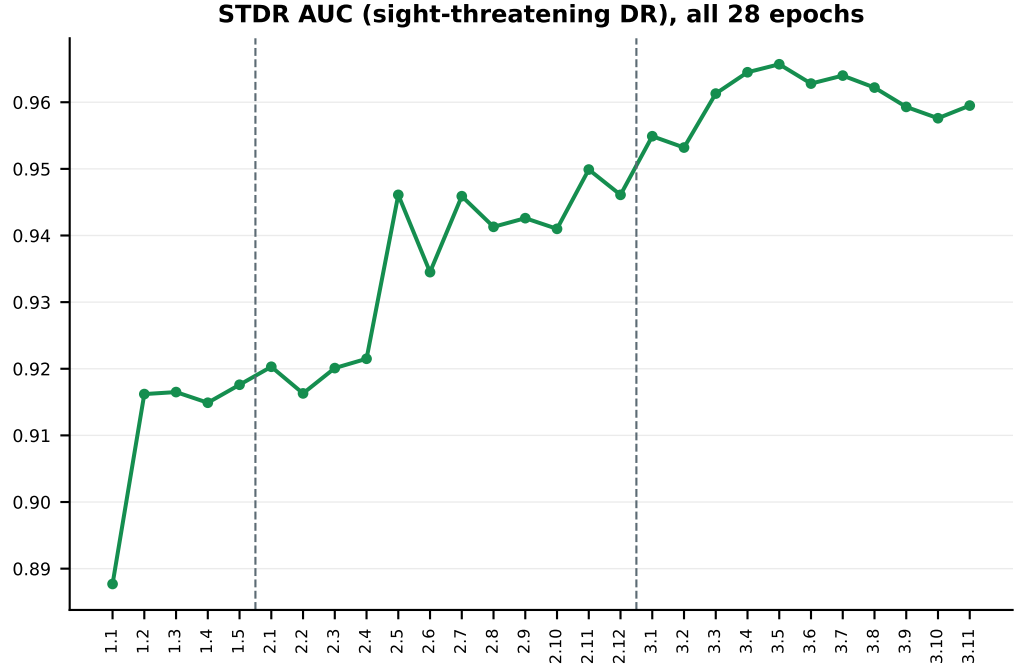


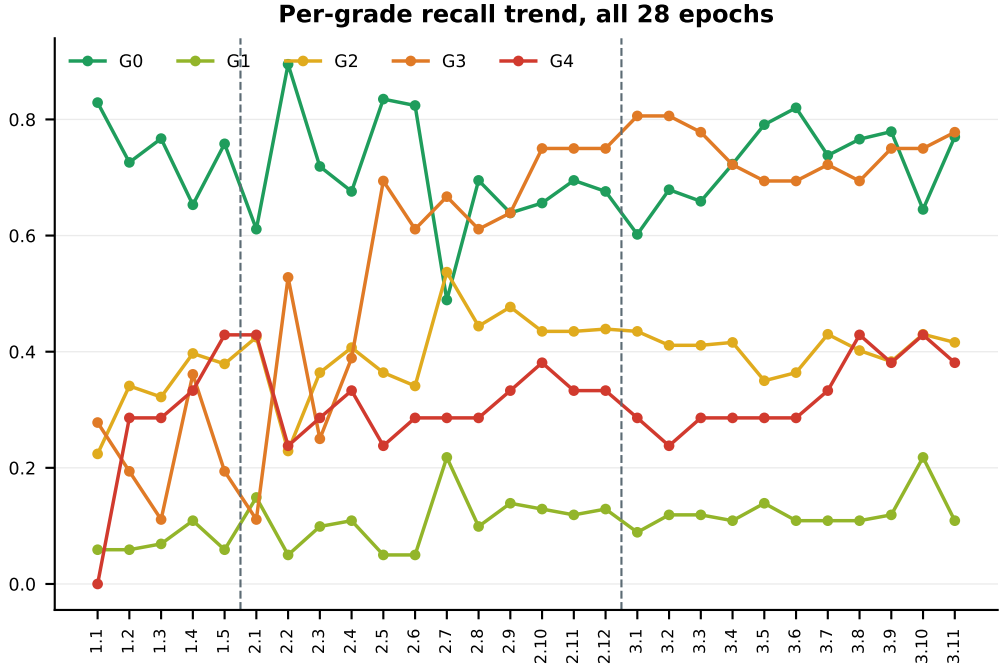


Trends


QWK & Score, all 28 epochs
QWK
Score
0.60
0.55
0.50
0.45
0.40
0.35
0.30
1.1 1.2 1.3 1.4 1.5 2.1 2.2 2.3 2.4 2.5 2.6 2.7 2.8 2.9 2.10 2.11 2.12 3.1 3.2 3.3 3.4 3.5 3.6 3.7 3.8 3.9 3.10 3.11


Loss, all 28 epochs
80
70
60
50
40
30
20
10
1.1 1.2 1.3 1.4 1.5 2.1 2.2 2.3 2.4 2.5 2.6 2.7 2.8 2.9 2.10 2.11 2.12 3.1 3.2 3.3 3.4 3.5 3.6 3.7 3.8 3.9 3.10 3.11


STDR AUC (sight-threatening DR), all 28 epochs
0.96
0.95
0.94
0.93
0.92
0.91
0.90
0.89
1.1 1.2 1.3 1.4 1.5 2.1 2.2 2.3 2.4 2.5 2.6 2.7 2.8 2.9 2.10 2.11 2.12 3.1 3.2 3.3 3.4 3.5 3.6 3.7 3.8 3.9 3.10 3.11

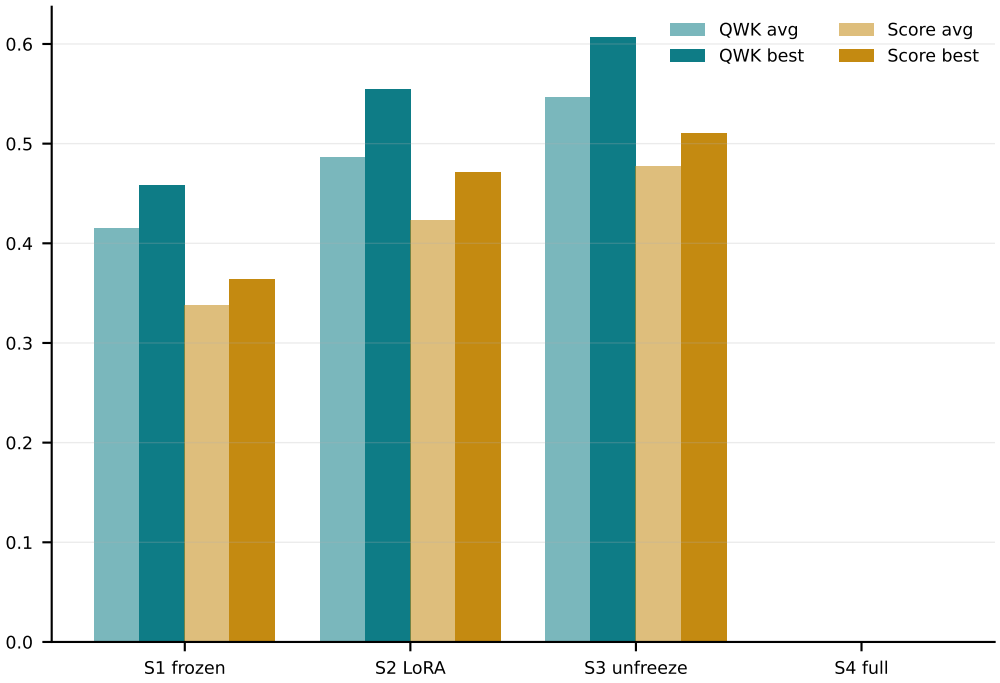

Per-grade recall trend, all 28 epochs
G0
G1
G2
G3
G4
0.8
0.6
0.4
0.2
0.0
1.1 1.2 1.3 1.4 1.5 2.1 2.2 2.3 2.4 2.5 2.6 2.7 2.8 2.9 2.10 2.11 2.12 3.1 3.2 3.3 3.4 3.5 3.6 3.7 3.8 3.9 3.10 3.11

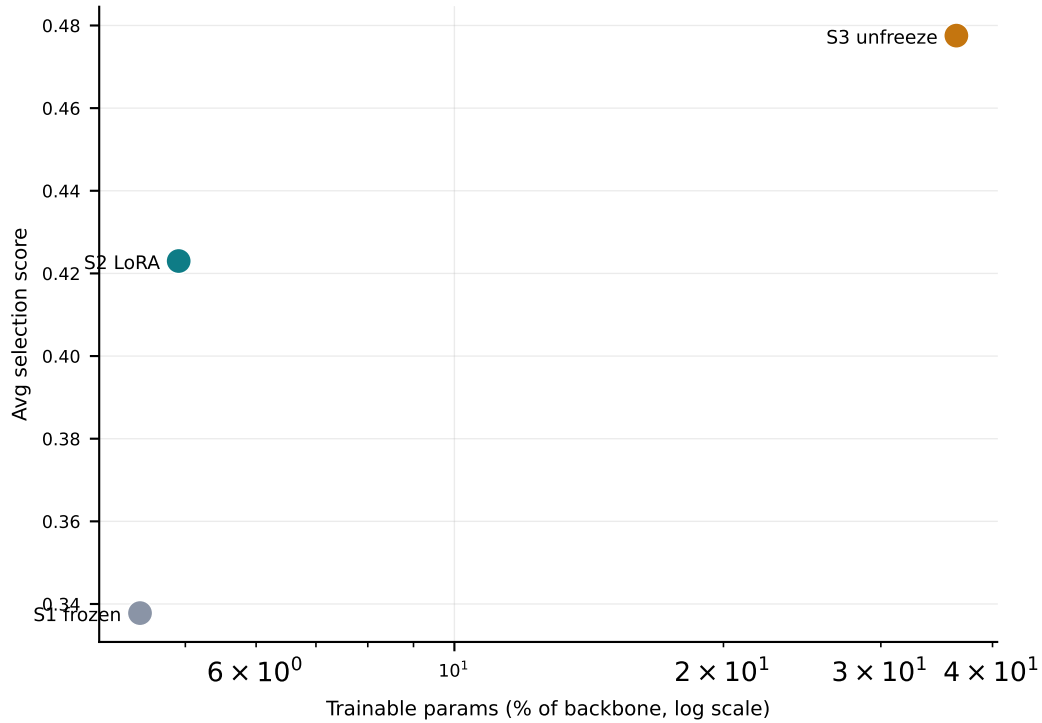
Stage-by-Stage Comparison

What each stage changed vs. what it achieved - S4 is config only, not yet run.

Stage	Epochs	Trainable params	% of backbone	Head / ladder LR	LoRA / backbone LR	Avg QWK	Best QWK	Avg score	Best score	Avg refAUC	Avg min-rec	Avg min /epoch
S1 frozen	5/5	14,220,579	4.45%	1.0e-4 / 1.0e-3	n/a / n/a	0.415	0.458	0.338	0.364	0.781	0.211	73
S2 LoRA	12/12	15,709,051	4.92%	1.0e-4 / 1.0e-3	5.0e-5 / n/a	0.486	0.554	0.423	0.471	0.830	0.361	93
S3 unfreeze	11/14 (running)	116,478,843	36.44%	5.0e-5 / 5.0e-4	5.0e-5 / 5.0e-6	0.547	0.607	0.478	0.510	0.862	0.394	164
S4 full	0/14	319,612,795	100.00%	2.0e-5 / 2.0e-4	2.0e-5 / 5.0e-6	-	-	-	-	-	-	-

Each capacity jump has paid off so far (avg QWK 0.415->0.486->0.543); S3's avg epoch time is skewed low by one resume-split epoch (true cost closer to 195-300 min) - see page 8.





Notable events

Grade 1 (Mild) breaking its floor - then matching its ceiling

Stuck at 5-11% recall for 12 straight epochs, Mild broke through from stage 2 epoch 7 onward, then tied its all-time-best 21.8% again at stage 3 epoch 10 - the first repeat of that ceiling in 18 epochs, landing on the same epoch QWK/score dipped (see page 9).

Two resume bugs found this week - both now fixed and confirmed

Mid-epoch resume (added Sep 29) missed its own first test (stale in-memory code); a Sep 30 retry confirmed it working. Separately, ONE patience counter spanned every stage, so stage 2's plateau silently killed stage 3 after 1 epoch - fixed to reset per stage; resumed into epoch 2.

Stage 3 breaks the plateau the decision bar was watching for

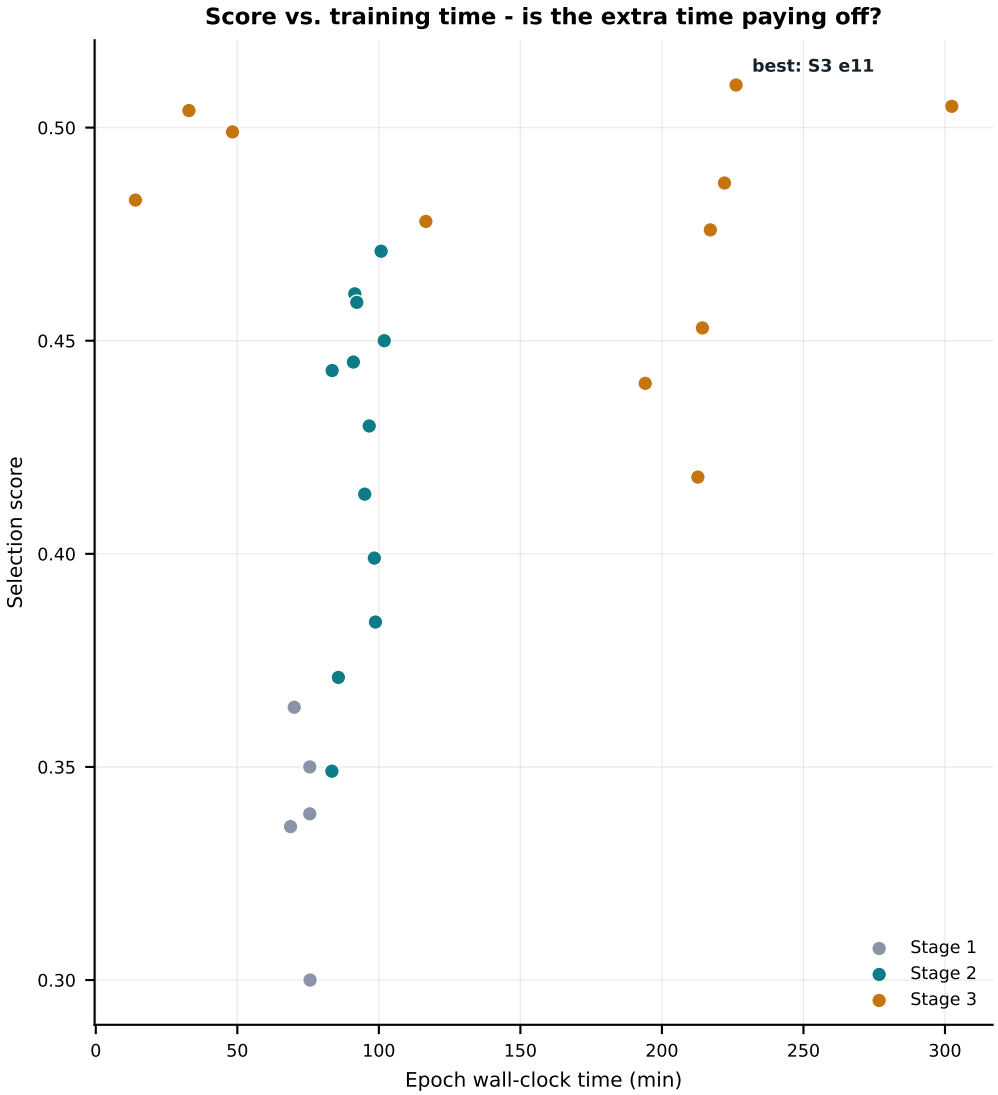
Epoch 1 regressed (score 0.418) as expected, but by epoch 4 it had already passed the old best (0.471, stage 2 epoch 5) and kept climbing - QWK 0.493->0.607, score 0.450->0.505, refAUC 0.840->0.873, all new run highs at stage 3 epoch 6. Unfreezing the backbone worked.

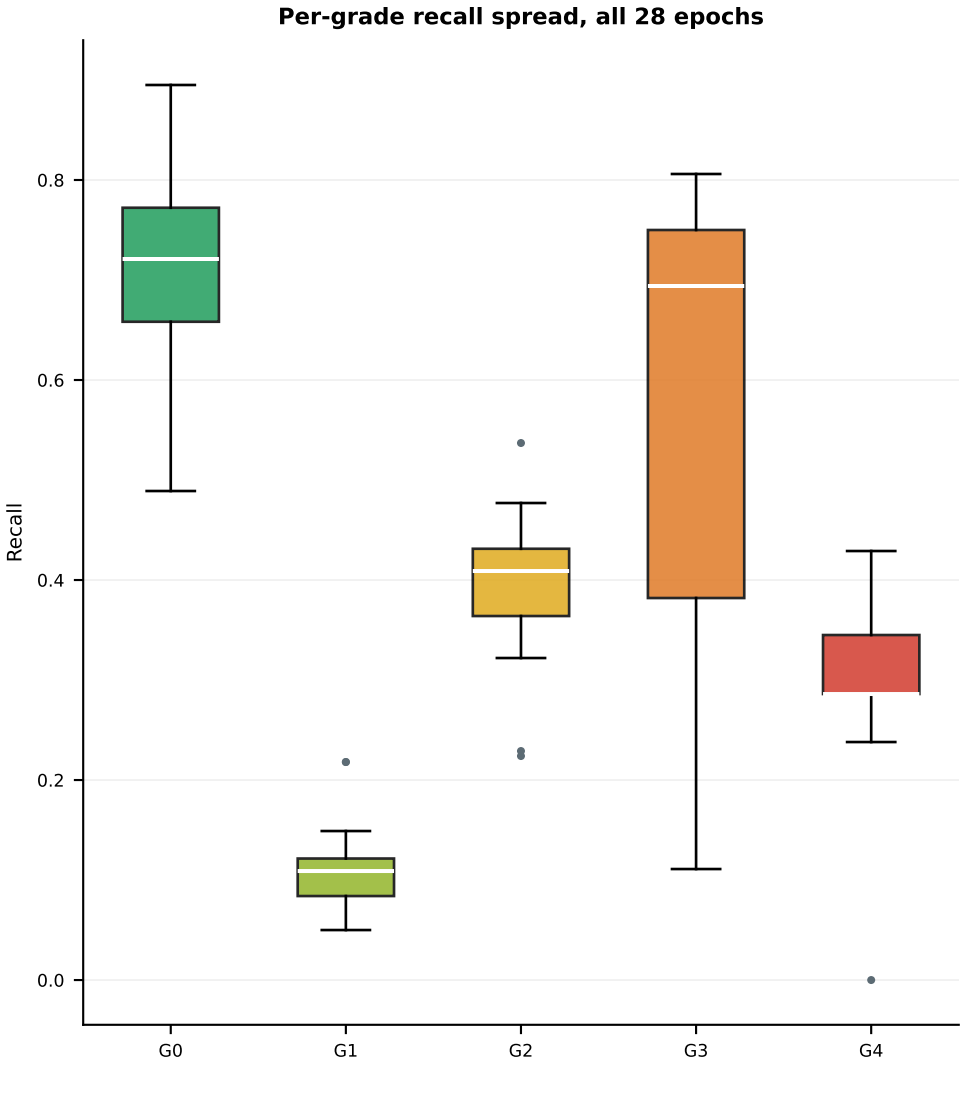
Setup & architecture reference

Dataset	35,126 images, 17,563 patients (EyePACS)
Split	28,102 / 3,512 / 3,512 train/val/test, no patient leakage
RETFound coverage	100% - 294/294 backbone tensors loaded
GLA-LoRA params	1,431,552 - 0.45% of the 319.6M backbone
Sampling ratio	5.01x grade4:grade0 weight, n^-0.5

Efficiency & Spread

Two more views of the same 28 epochs: is the extra training time paying off, and how much does each grade's recall vary epoch to epoch.





Detailed Analysis & Conclusion

The full read on 28 epochs, written out rather than just tabulated (1 of 2).

1. Trajectory

Three stages, three distinct regimes, each measurably better than the last. Stage 1 (head only, backbone frozen) established a floor: avg QWK 0.415, avg score 0.338 - the model learns the task at all, but with almost no capacity to adapt its visual features to fundus images. Stage 2 (GLA-LoRA adapters, still-frozen backbone) lifted every average metric (QWK to 0.486, score to 0.423) using only 1.43M extra parameters (0.45% of the backbone) and then plateaued hard for its last 7 epochs - epochs 11-17 never beat epoch 10's score of 0.471. Stage 3 (33% of the backbone blocks unfrozen) broke that plateau decisively: by epoch 4 it had already passed stage 2's best, and by epoch 6 it reached QWK 0.607 and score 0.505 - the run's best result to date, on every headline metric at once (QWK, score, refAUC, STDR AUC).

2. Why stage 3 worked

The stage 2 plateau was not a sign the model had learned everything it could from this data - it was a sign that 0.45%-of-backbone adapter capacity, with the backbone itself permanently frozen, had run out of room to represent whatever stage 1/2 hadn't already captured. Unfreezing even a third of the backbone (36.4% of all parameters now trainable, at a deliberately low 5e-6 backbone LR to avoid catastrophic forgetting of the RETFound pretraining) gave the model new degrees of freedom specifically in the deeper visual representation, not just in a lightweight adapter sitting on top of it. Epoch 1 of stage 3 regressed on every metric (score 0.450->0.418) - the same destabilization pattern seen at the stage 1->2 boundary - before recovering and then exceeding the prior ceiling, which is the expected signature of a genuine capacity increase rather than overfitting to validation noise.

3. What's still weak

Grade 1 (Mild NPDR) recall remains the model's clearest weakness: it averaged 12.4% across all 10 stage-3 epochs so far, nowhere near grade 0's 60-82% or grade 3's 69-81%. This almost certainly reflects the training split's own composition - grade 1 sits between the overwhelming grade-0 majority and the more visually distinct grade-2/3 lesions, and is the easiest grade for the ordinal head to blur into its neighbours. Notably, epoch 10 tied the run's all-time-best Mild recall (21.8%, exactly matching stage 2 epoch 7) - the first time that ceiling has been touched again in 18 epochs, and it happened on an epoch whose QWK/score both dropped, with grade 4 recall simultaneously at its joint-best (42.9%) too. That pattern - minority-grade recall up, headline ordinal agreement down, in the same epoch - reads as the model trading some majority-grade precision for minority-grade sensitivity, not a random bad epoch.

Detailed Analysis & Conclusion

The full read on 28 epochs, written out rather than just tabulated (2 of 2).

4. Deployment-relevant safety signal

Separately from grading accuracy, the screening gate (flag for human review vs. clear automatically, calibrated to a 0.985 negative-predictive-value target) has held in every one of the 28 epochs with no exception, including stage 3's regressed epoch 1. This is a different and arguably more deployment-critical claim than QWK: it says that even while the model's grading quality was fluctuating, the triage layer sitting in front of it was not sending disease-positive cases home undetected. That constraint being stage-transition-proof is a meaningfully separate result from the grading-quality trajectory above.

5. Risk & outlook

Epoch 7's dip (score 0.505->0.483) was followed by two epochs of recovery (0.499, then 0.504), then a second and deeper dip at epoch 10 (0.478 - the lowest score anywhere in stage 3 after epoch 1's initial regression). Two dips in ten epochs, with the best point still three epochs back at epoch 6, is a weaker trend than one dip would have been - but epoch 10's dip came paired with a tie for the best-ever Mild recall and joint-best PDR recall, which is a different and more encouraging story than a plain regression. The open question for the remaining 4 stage-3 epochs and all of stage 4 is whether QWK/score resume climbing (capacity was still the binding constraint and epoch 10 was a minority-recall-favouring fluctuation), settle into a noisier plateau around epoch 6-10's range, or - unchanged from the prior read - the minority-grade ceiling is closer to data-limited than capacity-limited, in which case stage 4's full unfreeze would mainly buy marginal gains on the majority grades.

6. Conclusion & recommendation

The run has already answered its central open question as of the last update: unfreezing backbone capacity was worth the ~2x epoch-time cost, decisively. best.pt still points to stage 3 epoch 6, not stage 2 epoch 10 (the old best). Recommendation: let stage 3 finish its remaining epochs and let stage 4 begin - there is no evidence yet of a ceiling that would justify stopping early - while watching two things specifically: whether QWK/score recover from epoch 10's dip the way they did from epoch 7's, and whether grade 1 recall's new tie at 21.8% becomes a genuine new floor or stays a one-epoch spike - the clearest open question this report cannot yet answer.